Solutions – Practice Problem Set 4

ECON 320 – Introduction to Mathematical Economics

Fall 2025

Optimization of Multivariable Functions (Chapter 11)

1. Key definitions

Critical point: A point x^* where $\nabla f(x^*) = \mathbf{0}$.
Gradient: $\nabla f = (\partial f/\partial x_1, \ldots, \partial f/\partial x_n)^\top$ (direction of steepest ascent).
Hessian: $H(x) = \nabla^2 f(x)$, the matrix of second partials.
Definiteness: A symmetric matrix H is PD/ND/indefinite according to the sign of the quadratic form $v^\top H v$ for all nonzero v (or via Sylvester's criterion using leading principal minors).

2. Classification in $\mathbb{R}^2$ via Hessian Let $H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$ at a stationary point. With $D = \det(H) = f_{xx}f_{yy} - f_{xy}^2$:

$$\begin{cases} D > 0,\ f_{xx} > 0 \ \Rightarrow\ \text{local minimum;} \\ D > 0,\ f_{xx} < 0 \ \Rightarrow\ \text{local maximum;} \\ D < 0 \ \Rightarrow\ \text{saddle;} \\ D = 0 \ \Rightarrow\ \text{inconclusive.} \end{cases}$$

3. $f(x, y) = 4x^2 + y^2 - 4xy + 2x - 3y$.

Step 1: FOCs.

$$f_x = 8x - 4y + 2, \quad f_y = 2y - 4x - 3.$$

Set to zero:

$$\begin{cases} 8x - 4y + 2 = 0 \ \Rightarrow\ 4x - 2y = -1, \\ 2y - 4x - 3 = 0 \ \Rightarrow\ 4x - 2y = -3. \end{cases}$$

Inconsistent $\Rightarrow$ *no stationary point.*

Step 2: Second derivatives.

$$H = \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix}, \quad \det(H) = 8 \cdot 2 - (-4)^2 = 16 - 16 = 0.$$

Singular Hessian indicates a flat direction in the quadratic form; combined with linear terms, the gradient system is inconsistent, hence no critical point (the function is unbounded along feasible directions).

4. $f(x, y) = 3x^2y - y^3 - 12x^2 + 12y$.

Step 1: FOCs.

$$f_x = 6xy - 24x = 6x(y - 4), \qquad f_y = 3x^2 - 3y^2 + 12.$$

So $f_x = 0 \Rightarrow x = 0$ or $y = 4$.

Case A: $x = 0$. Then $f_y = -3y^2 + 12 = 0 \Rightarrow y = \pm 2$. Points: $(0, 2)$ and $(0, -2)$.

Case B: $y = 4$. Then $f_y = 3x^2 - 48 + 12 = 3x^2 - 36 = 0 \Rightarrow x = \pm 2\sqrt{3}$. Points: $(2\sqrt{3}, 4)$ and $(-2\sqrt{3}, 4)$.

Step 2: Hessian and classification.

$$f_{xx} = 6y - 24, \quad f_{xy} = 6x, \quad f_{yy} = -6y, \quad D = f_{xx}f_{yy} - f_{xy}^2.$$

Evaluate:

- $(0, 2)$: $f_{xx} = -12$, $f_{yy} = -12$, $f_{xy} = 0 \Rightarrow D = 144 > 0$ and $f_{xx} < 0 \Rightarrow$ **local maximum**.
- $(0, -2)$: $f_{xx} = -36$, $f_{yy} = 12$, $f_{xy} = 0 \Rightarrow D = -432 < 0 \Rightarrow$ **saddle**.
- $(\pm 2\sqrt{3}, 4)$: $f_{xx} = 0$, $f_{yy} = -24$, $f_{xy} = \pm 12\sqrt{3} \Rightarrow D = -432 < 0 \Rightarrow$ **saddles**.

5. $f(x, y, z) = 2x^2 + y^2 + z^2 - 2xy - 4xz + 6yz - 3x + y + 2z$.

Step 1: FOCs.

$$\begin{cases} f_x = 4x - 2y - 4z - 3 = 0, \\ f_y = 2y - 2x + 6z + 1 = 0, \\ f_z = 2z - 4x + 6y + 2 = 0. \end{cases}$$

Solve:

$$(x^*, y^*, z^*) = \left(\frac{17}{18}, \frac{5}{18}, \frac{1}{18}\right).$$

Step 2: Hessian and classification.

$$H = \begin{bmatrix} 4 & -2 & -4 \\ -2 & 2 & 6 \\ -4 & 6 & 2 \end{bmatrix}, \quad \det(H) = -72 < 0.$$

Negative determinant for a 3×3 symmetric Hessian $\Rightarrow$ **indefinite** $\Rightarrow$ **saddle point** at the stationary point.

6. $Q(x, y) = x^{1/2}y^{1/2}$.

Marginal products.

$$MP_x = \frac{\partial Q}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2}, \quad MP_y = \frac{\partial Q}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2}.$$

Diminishing marginal returns.

$$\frac{\partial^2 Q}{\partial x^2} = -\frac{1}{4}x^{-3/2}y^{1/2} < 0, \quad \frac{\partial^2 Q}{\partial y^2} = -\frac{1}{4}x^{1/2}y^{-3/2} < 0.$$

Returns to scale. For $t > 0$, $Q(tx, ty) = t^{1/2}t^{1/2}Q(x, y) = tQ(x, y) \Rightarrow$ **constant returns to scale**.

7. Profit maximization.

(a) One product. $p = 100 - 2Q$, $C(Q) = 10Q + \frac{1}{2}Q^2$.

$$\pi(Q) = pQ - C(Q) = (100 - 2Q)Q - (10Q + \tfrac{1}{2}Q^2) = 90Q - \tfrac{5}{2}Q^2.$$

FOC: $\pi'(Q) = 90 - 5Q = 0 \Rightarrow Q^* = 18$. SOC: $\pi''(Q) = -5 < 0 \Rightarrow$ **maximum**.

(b) Two products.

$$\pi(x, y) = 100x + 80y - x^2 - y^2 - xy - 20(x + y).$$

FOCs:

$$\pi_x = 80 - 2x - y = 0, \qquad \pi_y = 60 - 2y - x = 0.$$

Solve:

$$2x + y = 80, \quad x + 2y = 60 \;\Rightarrow\; (x^*, y^*) = \left(\frac{100}{3}, \frac{40}{3}\right).$$

Hessian:

$$H = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}.$$

Eigenvalues -3 and $-1 \Rightarrow$ **negative definite** $\Rightarrow$ **strict global maximum**.

Optimization with Equality Constraints (Chapter 12)

8. Economic meaning of the multiplier λ

In $\max f(x)$ s.t. $g(x) = \bar{c}$, the Lagrange multiplier λ^* equals the *shadow value*:

$$\lambda^* = \frac{\partial \mathcal{L}}{\partial \bar{c}} = \frac{\partial f^*}{\partial \bar{c}},$$

the marginal improvement in the optimal objective value when the constraint is relaxed by one unit.

- *Utility maximization (budget):* λ is the marginal utility of income.
- *Cost minimization (output target):* λ is the marginal cost of relaxing the output requirement (i.e., Lagrangean shadow price on the constraint).

9. Second-order conditions (bordered Hessian)

With m equality constraints, bordered Hessian

$$H_b = \begin{bmatrix} 0 & G \\ G^\top & H \end{bmatrix}, \quad G = \nabla g(x^*) \in \mathbb{R}^{m\times n},\ H = \nabla^2_{xx}\mathcal{L}(x^*, \lambda^*).$$

For the highest-order leading principal minor (of full H_b):

$$(-1)^m \det(H_b) \begin{cases} < 0 & \text{maximum,} \\ > 0 & \text{minimum.} \end{cases}$$

Intuition: the border flips sign m times because we test curvature *restricted to the tangent space* of the constraints.

10. $\max f(x, y) = xy$ s.t. $x + y = 10$.

Step 1: Lagrangian & FOCs.

$$\mathcal{L} = xy - \lambda(x + y - 10), \quad \mathcal{L}_x = y - \lambda = 0,\ \mathcal{L}_y = x - \lambda = 0,\ x + y = 10.$$

Thus $x = y = \lambda$ and $2x = 10 \Rightarrow (x^*, y^*) = (5, 5)$.

Step 2: SOC via restriction or border. On the line $y = 10 - x$, $f(x) = x(10 - x) = 10x - x^2$ with $f''(x) = -2 < 0 \Rightarrow$ **maximum**.
Bordered Hessian:

$$H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},\ G = \begin{bmatrix} 1 & 1 \end{bmatrix},\ H_b = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$\det(H_b) = 2 > 0$. For $m = 1$, $(-1)^m \det(H_b) = -2 < 0 \Rightarrow$ **maximum**.

11. $\max Q = 4\sqrt{xy}$ s.t. $2x + y = 100$ (maximize output given a cost).

Step 1: FOCs.

$$\mathcal{L} = 4\sqrt{xy} - \lambda(2x + y - 100).$$

$$\mathcal{L}_x = \frac{2y}{\sqrt{xy}} - 2\lambda = 0 \;\Rightarrow\; \lambda = \sqrt{\frac{y}{x}}, \quad \mathcal{L}_y = \frac{2x}{\sqrt{xy}} - \lambda = 0 \;\Rightarrow\; \lambda = 2\sqrt{\frac{x}{y}}.$$

Equate: $\sqrt{y/x} = 2\sqrt{x/y} \Rightarrow y/x = 4x/y \Rightarrow y = 2x$. Constraint: $2x + y = 100 \Rightarrow 2x + 2x = 100 \Rightarrow x^* = 25,\ y^* = 50$.

Step 2: SOC. $4\sqrt{xy}$ is concave on $\mathbb{R}^2_{++}$ and the constraint is affine $\Rightarrow$ **global maximum**. (If one insists on a second-derivative check: restrict to $y = 100 - 2x$, consider $\phi(x) = 4\sqrt{x(100 - 2x)}$; then $\phi''(25) < 0$.)

Step 3: Economic meaning. $\lambda^* = \sqrt{y/x} = \sqrt{2}$ is the marginal increase in maximal Q per marginal increase in the expenditure cap.

12. $\min f = x^2 + y^2 + z^2$ s.t. $x + y + z = 3$.

Step 1: FOCs.

$$\mathcal{L} = x^2 + y^2 + z^2 - \lambda(x + y + z - 3), \quad 2x - \lambda = 0,\ 2y - \lambda = 0,\ 2z - \lambda = 0.$$

Thus $x = y = z = \lambda/2$. Constraint $\Rightarrow 3(\lambda/2) = 3 \Rightarrow \lambda = 2$. Hence $(x^*, y^*, z^*) = (1, 1, 1)$.

Step 2: SOC. f is strictly convex; affine constraint $\Rightarrow$ unique **global minimum**. (For the bordered Hessian, $H = 2I_3$, $G = [1, 1, 1]$, $\det(H_b) = -12 < 0$; with $m = 1$, $(-1)^m \det(H_b) > 0$ holds for a minimum.)

13. $f = x^2 + y^2$ s.t. $x + 2y = 4$.

Step 1: FOCs.

$$\mathcal{L} = x^2 + y^2 - \lambda(x + 2y - 4),\ 2x - \lambda = 0,\ 2y - 2\lambda = 0.$$

Thus $x = \lambda/2,\ y = \lambda$. Constraint $\Rightarrow \lambda/2 + 2\lambda = 4 \Rightarrow \frac{5}{2}\lambda = 4 \Rightarrow \lambda = \frac{8}{5}$. Hence $(x^*, y^*) = (\frac{4}{5}, \frac{8}{5})$.

Step 2: SOC & existence. f is strictly convex, constraint affine $\Rightarrow$ unique **global minimum**. No maximum (unbounded above along the line).

14. $U(x, y) = x^{0.4}y^{0.6}$, budget $4x + 2y = 100$.

Step 1: FOCs (MRS = price ratio).

$$\frac{MU_x}{MU_y} = \frac{0.4\,x^{-0.6}y^{0.6}}{0.6\,x^{0.4}y^{-0.4}} = \frac{2}{3} \cdot \frac{y}{x} = \frac{p_x}{p_y} = \frac{4}{2} = 2 \;\Rightarrow\; \frac{2}{3} \cdot \frac{y}{x} = 2 \;\Rightarrow\; y = 3x.$$

Budget: $4x + 2(3x) = 10x = 100 \Rightarrow x^* = 10,\ y^* = 30$.

Step 2: SOC. Cobb–Douglas utility is strictly concave on $\mathbb{R}^2_{++}$; budget is affine $\Rightarrow$ **global maximum**. (Bordered-Hessian or second-derivative along the budget line would confirm.)

15. Identity: $\det(H_b) = \det(H)\,\det(-GH^{-1}G^\top)$ (when H is nonsingular)

Let $A = 0_{m\times m}$, $B = G$, $C = G^\top$, $D = H$. By the block determinant formula (Schur complement),

$$\det\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D)\,\det\big(A - BD^{-1}C\big) = \det(H)\,\det\big(0 - GH^{-1}G^\top\big).$$

Thus $\det(H_b) = \det(H)\,\det(-GH^{-1}G^\top)$. Interpretation: $-GH^{-1}G^\top$ is the (negative) *projected curvature* of $\mathcal{L}$ on the constraint normals; its sign determines concavity/convexity in feasible directions.

16. $\min f = x^2 + y^2 + z^2$ s.t. $\begin{cases} x + y + z = 3, \\ x - y + 2z = 4. \end{cases}$

Step 1: Lagrangian & FOCs.

$$\mathcal{L} = x^2 + y^2 + z^2 - \lambda_1(x + y + z - 3) - \lambda_2(x - y + 2z - 4).$$

$$2x - \lambda_1 - \lambda_2 = 0, \quad 2y - \lambda_1 + \lambda_2 = 0, \quad 2z - \lambda_1 - 2\lambda_2 = 0.$$

Step 2: Solve with constraints. From the three FOCs:

$$x = \tfrac{1}{2}(\lambda_1 + \lambda_2), \quad y = \tfrac{1}{2}(\lambda_1 - \lambda_2), \quad z = \tfrac{1}{2}(\lambda_1 + 2\lambda_2).$$

Plug into constraints:

$$\begin{cases} x + y + z = \tfrac{3}{2}\lambda_1 + \tfrac{1}{2}\lambda_2 = 3, \\ x - y + 2z = \tfrac{1}{2}\lambda_1 + \tfrac{7}{2}\lambda_2 = 4. \end{cases}$$

Solve:

$$\lambda_1 = \tfrac{10}{7}, \quad \lambda_2 = \tfrac{6}{7}.$$

Hence

$$x^* = \tfrac{8}{7}, \quad y^* = \tfrac{2}{7}, \quad z^* = \tfrac{11}{7}.$$

Step 3: SOC. f is strictly convex, constraints affine $\Rightarrow$ unique **global minimum**.

(If desired, bordered-Hessian check: with $H = 2I_3$, $G = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$, $\det(H_b) = \det(H)\det(-GH^{-1}G^\top) = 8 \cdot \det\big(-\tfrac{1}{2}GG^\top\big) = 8 \cdot \tfrac{1}{4}\det(GG^\top) = 8 \cdot \tfrac{1}{4} \cdot 14 = 28 > 0$; for $m = 2$, $(-1)^m \det(H_b) > 0$ confirms a minimum.)